

Preclinical Target Assessment and Small-Molecule Discovery Strategies for the IL-23p19 / IL-23R Interaction in Plaque Psoriasis

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Abstract

Interleukin-23 (IL-23) is the validated master cytokine driver of pathogenic T helper 17 (Th17) cell-mediated inflammation in moderate-to-severe plaque psoriasis. Although monoclonal antibodies targeting the specific IL-23p19 subunit (e.g., guselkumab, risankizumab) achieve unprecedented clinical remission rates (PASI-90 > 70%), their parenteral administration, high manufacturing cost, and immunogenicity underscore a substantial unmet need for oral therapeutics. Oral peptide antagonists such as icotrokinra (JNJ-77242113) have established clinical proof-of-concept for non-antibody targeting of the IL-23 receptor (IL-23R), but remain limited by low oral bioavailability (< 2%), macrocyclic synthetic complexity, and pharmacokinetic variability. Conversely, the discovery of orally bioavailable, rule-of-5-compliant small molecules directly disrupting the IL-23p19 / IL-23R protein-protein interaction (PPI) has historically been impeded by the extensive, flat nature of the contact surface. Here, we present a comprehensive structural, mechanistic, and translational target evaluation synthesizing recent crystallographic breakthroughs—most notably the resolution of an induced-fit orthosteric pocket on IL-23p19 (PDB 8UUI) with substantial loop rearrangements (C α displacements up to 3.9 Å; all-atom RMSDs up to 6.0 Å). We delineate multi-pronged small-molecule discovery workflows spanning fragment-based drug discovery (FBDD), peptide displacement-based high-throughput screening (HTS), DNA-encoded library (DEL) counter-selection, and structure-based virtual screening. Furthermore, we define a 4-subpocket pharmacophore blueprint and establish a detailed Target Product Profile (TPP) for a preclinical small-molecule development candidate designed for once-daily oral administration in psoriasis.

Keywords: Interleukin-23, IL-23R, IL-23p19, Plaque Psoriasis, Protein-Protein Interaction (PPI), Induced Orthosteric Pocket, Small Molecule, Target Product Profile (TPP).

1 Introduction and Biological Rationale in Psoriasis

Plaque psoriasis is a chronic, immune-mediated inflammatory skin disorder affecting 2–3 % of the global population, characterized by well-demarcated, erythematous plaques covered with silvery scales [10]. The disease pathophysiology is governed by the IL-23 / Th17 immune signaling axis, wherein antigen-presenting dendritic cells release IL-23 to stimulate the differentiation, survival, and pathogenic maintenance of Th17 lymphocytes, $\gamma\delta$ T cells, and type 3 innate lymphoid cells (ILC3s) [6].

1.1 Cytokine Architecture and Receptor Signaling

Interleukin-23 is a heterodimeric helical cytokine belonging to the IL-12 family, composed of two covalently linked subunits:

1. **IL-23p19 (IL-23 α , UniProt Q9NPF7)**: A four- α -helix bundle cytokine subunit unique to IL-23, which determines receptor specificity and therapeutic selectivity.
2. **IL-12/23p40 (IL-12 β , UniProt P29460)**: A shared, three-domain immunoglobulin-like subunit linked to p19 via an essential interchain disulfide bridge between Cys54 of p19 and Cys177 of p40.

Signal transduction occurs when the IL-23 heterodimer binds two transmembrane receptor chains on target effector leukocytes (Fig. 1):

- **IL-23R (UniProt Q5VWK5)**: The cognate, high-affinity receptor subunit that binds almost exclusively to the p19 subunit through its N-terminal immunoglobulin domain (D1 domain) [3].
- **IL-12R β 1 (UniProt P42701)**: The shared receptor subunit that binds to the shared p40 subunit.

Upon assembly of the quaternary signaling complex, IL-23R activates Janus kinase 2 (JAK2) while IL-12R β 1 activates tyrosine kinase 2 (TYK2). This triggers trans-phosphorylation and subsequent recruitment and phosphorylation of Signal Transducer and Activator of Transcription 3 (STAT3), as well as STAT4, STAT1, and STAT5 [4, 7]. Phosphorylated STAT3 (pSTAT3) homodimerizes, translocates into the nucleus, and activates the transcription of effector cytokines including IL-17A, IL-17F, IL-22, TNF- α , and CCL20. These factors act directly on epidermal keratinocytes, inducing hyperproliferation, aberrant differentiation (parakeratosis), expression of antimicrobial peptides (S100A7/psoriasin, S100A8/A9, β -defensin 2), and recruitment of neutrophils into the stratum corneum, creating Munro's microabscesses.

1.2 Clinical Validation and Limitations of Existing Modalities

Targeting the IL-23 pathway has yielded superior clinical efficacy compared to legacy anti-TNF- α or anti-IL-12/23p40 therapies (Table 1 and Fig. 2). Monoclonal antibodies specifically directed against IL-23p19 (guselkumab, risankizumab, tildrakizumab) achieve PASI-75 (Psoriasis Area and Severity Index $\geq$ 75% reduction) in 85–92 % of patients and PASI-90 in 70–80 % of patients at week 16 [8, 11].

Despite this therapeutic success, injectable biologics face significant clinical and commercial drawbacks:

- **Invasive administration**: Regular subcutaneous self-injections cause injection-site pain, needle phobia, and decreased long-term adherence.
- **Immunogenicity**: Development of anti-drug antibodies (ADAs) can lead to secondary loss of response over time.
- **Logistical and financial barriers**: Cold-chain storage requirements and steep annual acquisition costs (60 000–80 000 USD per patient) restrict global patient access.

The emergence of oral peptide antagonists, notably icotrokinra (JNJ-77242113 / PN-235), has demonstrated clinical proof-of-concept that targeting IL-23R outside of an antibody scaffold can achieve biologic-like efficacy (PASI-75 of 79 %, PASI-90 of 60 % in Phase 2/3 trials) [2, 5]. However, icotrokinra is a complex macrocyclic peptide of $\sim$ 1.5 kDa with an oral bioavailability below

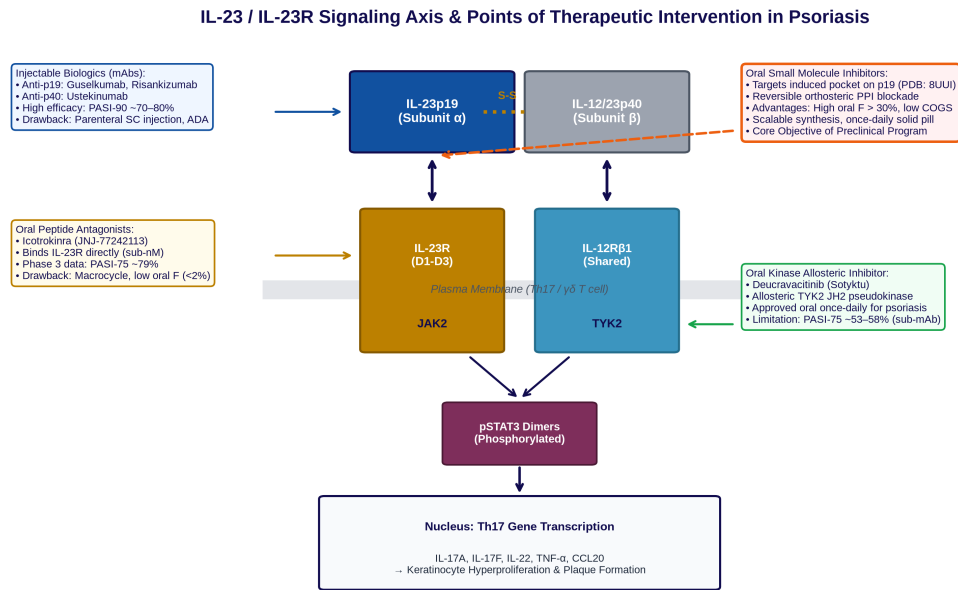


Figure 1. The IL-23 / IL-23R signaling axis and comparative points of therapeutic intervention in plaque psoriasis. The heterodimeric cytokine IL-23 binds cooperatively to cognate IL-23R and shared IL-12Rβ1. Receptor assembly triggers JAK2/TYK2 trans-phosphorylation and STAT3 dimerization, driving transcriptional activation of key pathogenic effector cytokines (IL-17A, IL-17F, IL-22). Highlighted callouts indicate therapeutic classes, including injectable mAbs, oral peptides, approved allosteric kinase inhibitors, and oral small-molecule orthosteric PPI inhibitors.

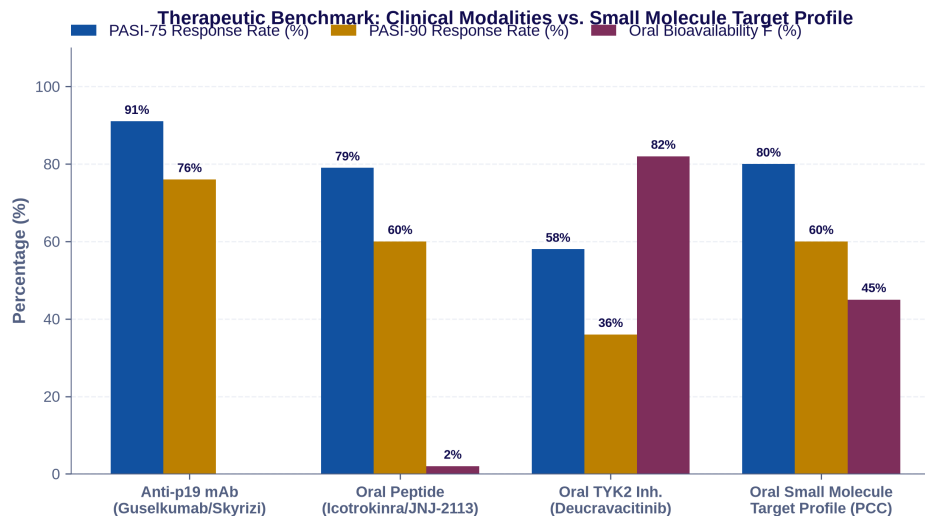


Figure 2. Quantitative therapeutic benchmark comparing clinical modalities targeting the IL-23 / Th17 axis in plaque psoriasis against the target profile for an oral small-molecule development candidate. While injectable mAbs provide high efficacy, they lack oral bioavailability. Oral peptides (icatrokinra) validate non-antibody IL-23R targeting with biologic-like efficacy, but suffer from low bioavailability (< 2%) and macrocyclic complexity. Approved oral TYK2 inhibitors (deucravacitinib) achieve lower clearing rates. An oral small molecule directly targeting the IL-23p19 / IL-23R interface combines biologic-level efficacy (PASI-75 ~ 80%) with high oral bioavailability (> 30%), room-temperature stability, and low cost of goods.

Table 1. Clinical and Preclinical Benchmarking of Therapeutic Modalities in Plaque Psoriasis.

Therapeutic Class	Representative	Target / Mechanism of Action	Route	PASI-75 (%)	PASI-90 (%)
Injectable mAb	Guselkumab	IL-23p19 neutralizing mAb	SC (q8w)	91	76
Injectable mAb	Risankizumab	IL-23p19 neutralizing mAb	SC (q12w)	91	75
Injectable mAb	Ustekinumab	IL-12/23p40 neutralizing mAb	SC (q12w)	76	44
Oral Peptide	Icotrokinra	IL-23R direct antagonist	Oral (QD/BID)	79	60
Oral Kinase Inh.	Deucravacitinib	TYK2 allosteric (JH2 pseudokinase)	Oral (QD)	58	36
Oral Small Molecule	Target Profile (PCC)	IL-23p19 orthosteric PPI inhibitor	Oral (QD)	≥ 75	≥ 55

2 %, requiring high doses (100 mg twice daily), specialized formulation technologies, and costly solid-phase chemical synthesis. On the other hand, the approved oral allosteric TYK2 inhibitor deucravacitinib achieves PASI-75 rates of only 53–58 % [1]. Hence, a small-molecule direct PPI inhibitor targeting the IL-23p19 / IL-23R interface represents the premier therapeutic goal: combining the efficacy and safety of selective IL-23p19 blockade with the convenience, high oral bioavailability, and low manufacturing cost of an oral solid dosage form.

2 Structural Biology of the IL-23p19 / IL-23R Interface

Understanding the quaternary architecture and interfacial residue interactions of the IL-23 signaling complex is indispensable for structure-based drug design. Seminal crystallographic and cryo-EM investigations (PDB entries [5MZV](#), [6WDQ](#), and [8OE4](#)) have resolved the molecular mechanism of cognate receptor recognition [3, 4, 7].

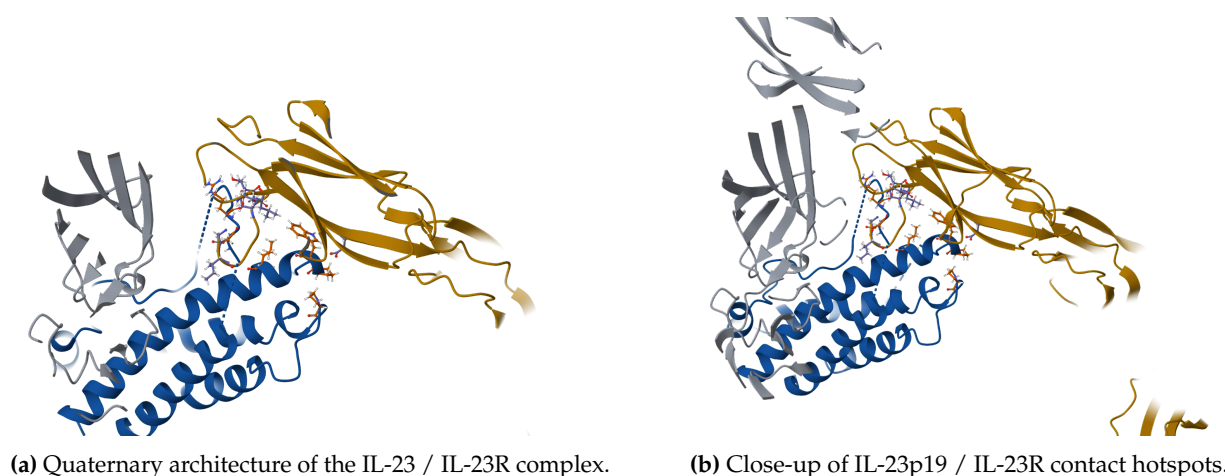


Figure 3. Structural biology of the human IL-23 / IL-23R complex (PDB [5MZV](#), 2.8 Å resolution). (a) The heterodimer IL-23 contains the specific p19 subunit (Acellera blue) and shared p40 subunit (gray), complexed with the cognate IL-23R N-terminal Ig-like domain (gold). (b) Detailed view of interfacial contact residues. Binding is mediated by key hydrogen bonds and salt bridges (p19 Arg57 to IL-23R Glu111/Thr112; p19 Glu59 to IL-23R Asn29) and hydrophobic packing (p19 Pro155, Trp156, Leu160 against IL-23R Asp118 and Leu113).

2.1 Complex Architecture and Hotspot Topology

IL-23 engages its cognate receptor IL-23R exclusively through its N-terminal immunoglobulin domain (D1 domain, spanning residues Gly24 to Asp120) [3]. The interaction buries approximately 1150 Å² of solvent-accessible surface area across helices α A and α D of IL-23p19 and the loop regions connecting them (Fig. 3).

Biophysical and structural analysis reveals that the p19 / IL-23R contact surface is dominated by discrete energetic hotspots (Table 2):

- **The Arg57–Glu111/Thr112 Salt Bridge / Polar Network:** Arg57 of p19 projects into a polar pocket on IL-23R, forming a bidentate hydrogen bonding and electrostatic interaction with the backbone carbonyl of Glu111 (3.20 Å) and the side-chain hydroxyl of Thr112 (2.82 Å). Mutational disruption of Arg57 abates IL-23R binding affinity by > 100-fold.

- **The Glu59–Asn29 Polar Pair:** Glu59 of p19 forms a critical hydrogen bond with Asn29 of IL-23R loop 1 (2.71 Å).
- **The Hydrophobic Anchor Pack (Pro155, Trp156, Leu160):** Helix α D of p19 presents a lipophilic ridge consisting of Trp156 and Leu160, flanked by Pro155, which packs against Leu113, Gly116, and the aliphatic stalk of Asp118 on IL-23R.
- **The Basic C-Terminal Anchor (Lys164):** The side-chain amino group of Lys164 forms an electrostatic contact with the backbone carbonyl of Gly24 on IL-23R (2.62 Å).

Table 2. Key Interfacial Residue Contacts at the IL-23p19 / IL-23R Interface (PDB [5MZV](#)).

IL-23p19 Residue	IL-23R Residue	Interaction Type	Distance (Å)	Structural Role
Arg57 (NH1)	Thr112 (OG1)	Hydrogen bond / Polar	2.82	Primary electrostatic anchor on loop 3
Arg57 (N)	Glu111 (O)	Backbone H-bond	3.20	Orientates p19 helix α A against D1 domain
Arg57 (O)	Leu113 (N)	Backbone H-bond	2.85	Stabilizes loop 3 conformation
Glu59 (O)	Asn29 (OD1)	Polar / H-bond	2.71	Key recognition contact on loop 1
Pro155 (CD)	Asp118 (OD2)	Hydrophobic / Polar	2.99	Packs adjacent to D1 hydrophobic core
Trp156 (CD1)	Asp118 (OD2)	Hydrophobic pack	3.15	Aromatic capping of interface
Leu160 (CD2)	Leu113 (CD1)	Lipophilic contact	3.65	Hydrophobic core stabilization
Lys164 (NZ)	Gly24 (O)	Salt bridge / Polar	2.62	C-terminal electrostatic anchor
Ser168 (CB)	Ile25 (CD1)	Van der Waals	3.16	Peripheral interface sealing

3 The Induced Orthosteric Pocket on IL-23p19: Druggability Breakthrough

Targeting cytokine-receptor interfaces with small molecules has historically been deemed high-risk due to the presumption of a featureless, flat contact topography. However, a landmark structural investigation by Lecomte et al. (PDB entry [8UUI](#), 2.43 Å resolution) established that the IL-23p19 interface possesses remarkable conformational plasticity, undergoing an induced-fit rearrangement upon ligand binding to reveal a deep, druggable orthosteric pocket [9].

3.1 Conformational Rearrangement and Pocket Generation

Comparative structural superposition of IL-23p19 in the IL-23R-bound conformation (PDB [5MZV](#)) versus the pocket-induced conformation (PDB [8UUI](#)) demonstrates substantial regional conformational mobility ([Fig. 5](#)). While the rigid core of the four-helix bundle remains stable (overall C α RMSD of 1.22 Å), two specific regions undergo striking conformational remodeling:

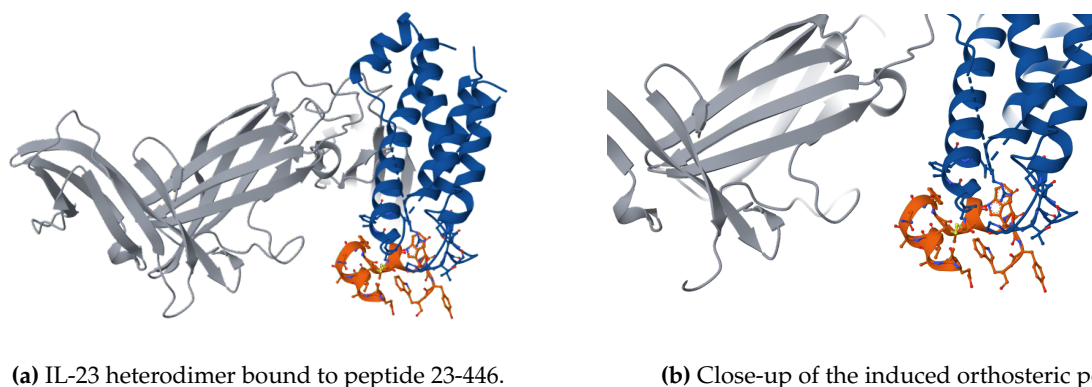


Figure 4. Crystallographic validation of the induced orthosteric pocket on IL-23p19 (PDB 8UUI, 2.43 Å resolution). (a) Overall structure of human IL-23 heterodimer (p19 in blue, p40 in gray) in complex with inhibitory peptide 23-446 (orange). (b) High-resolution view of the induced pocket on p19. The peptide aromatic side chains penetrate deeply into the induced hydrophobic cleft lined by Pro113, Ser114, Leu115, Leu116, Leu159, Leu160, and Phe163, providing a direct structural template for small-molecule design.

1. **Loop 3 Region (Residues 108–122):** Pro117 shifts by 3.29 Å ($C\alpha$), Asp118 shifts by 2.71 Å, and Pro113 shifts by 2.37 Å.
2. **Helix α D / C-Terminal Loop (Residues 153–165):** Pro155 exhibits a dramatic $C\alpha$ displacement of 3.93 Å (all-atom RMSD of 4.87 Å), while Gln154 undergoes a 3.39 Å $C\alpha$ shift (all-atom RMSD of 5.14 Å) and Gln157 shifts with an all-atom RMSD of 6.04 Å (Table 3).

This concerted displacement swings Pro155, Gln154, and Trp156 outward from the protein core, generating a concave hydrophobic cleft with an estimated volume of 400–500 Å³. This cavity is entirely occluded in the apo or receptor-bound cytokine structures.

3.2 Molecular Mode of Action of the Orthosteric Probe

In PDB 8UUI, the 18-mer peptide 23-446 occupies this induced pocket with high geometric and chemical complementarity (Fig. 4b):

- **Trp18 Insertion:** The indole ring of Trp18 buries deeply into the primary hydrophobic sub-pocket (S1), forming tight van der Waals contacts with Pro113, Leu115, Leu116, and Leu159 (closest contact 3.23 Å).
- **Tyr19 Recognition:** Tyr19 forms a critical backbone-to-backbone hydrogen bond between its amide nitrogen and the backbone carbonyl oxygen of Ser114 (2.75 Å), while its phenol ring stacks against Leu116 and Pro113. Additionally, the side chain of Ser119 forms a hydrogen bond with the peptide backbone (2.30 Å).
- **Trp20 and Phe14 Stacking:** Trp20 and Phe14 occupy the secondary lipophilic shelf (S3), packing against Leu160 and Phe163 (distances 3.48–3.50 Å).

Because this induced pocket directly overlaps the binding footprint of the IL-23R D1 domain, a small molecule occupying this site creates steric and electrostatic clashes that completely prohibit IL-23R engagement, thereby extinguishing downstream JAK2/STAT3 signaling.

Table 3. Conformational Displacement of Pocket-Lining Residues in IL-23p19 (5MZV vs. 8UUI).

Residue	C α Shift (Å)	All-Atom RMSD (Å)	Secondary Structure	Functional Role
Pro113	2.37	2.34	Loop 3	Forms S1 hydrophobic floor
Ser114	1.03	1.22	Loop 3	Backbone C=O H-bond to Tyr19
Leu115	0.90	1.89	Loop 3	Lipophilic wall of S1
Leu116	2.30	2.74	Loop 3	Encloses hydrophobic pocket
Pro117	3.29	3.78	Loop 3	Major conformational hinge
Asp118	2.71	2.84	Loop 3	Solvent-exposed polar margin
Ser119	1.44	1.38	Loop 3	Side-chain polar anchor
Gln154	3.39	5.14	Helix α D	Swings open to reveal cleft
Pro155	3.93	4.87	Helix α D	Primary induced-fit displacement
Trp156	1.67	1.78	Helix α D	Rotates side chain away
Gln157	2.13	6.04	Helix α D	Flexible polar peripheral cap
Leu159	0.85	1.81	Helix α D	Deep lipophilic ceiling of S1
Leu160	1.07	1.08	Helix α D	Forms S3 aromatic shelf
Arg162	0.64	1.07	Helix α D	Basic side chain at solvent exit
Phe163	0.62	1.86	Helix α D	Hydrophobic wall of S3

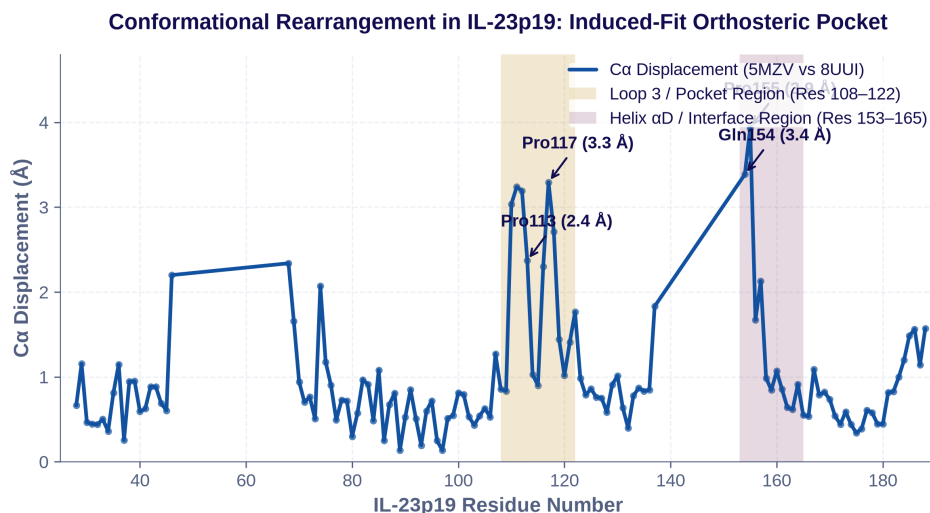


Figure 5. Quantitative C α displacement profile of IL-23p19 across 124 aligned common residues between the cognate IL-23R-bound state (PDB 5MZV) and the induced-pocket state (PDB 8UUI). The analysis highlights two localized regions of substantial structural remodeling: the loop 3 / pocket margin (residues 108–122, gold shading) and the helix α D interface region (residues 153–165, plum shading). Displacements reaching 3.93 Å for Pro155 and 3.39 Å for Gln154 open the cryptic hydrophobic cavity.

4 Small-Molecule Discovery Strategies for the IL-23p19 / IL-23R Interaction

Achieving potent small-molecule inhibition of this cytokine-receptor PPI requires an integrated screening and biophysical validation campaign (Fig. 6). Several complementary discovery paradigms can be deployed to yield drug-like chemical matter.

4.1 Primary Hit Identification Modalities

4.1.1 Fragment-Based Drug Discovery (FBDD)

Fragment-based screening is uniquely well-suited for interrogating induced pockets on protein interfaces.

- **Library Design:** A curated collection of 2000–3000 fragments conforming to the “Rule of Three” ($MW \leq 300$ Da, $cLogP \leq 3$, $HBD \leq 3$, $HBA \leq 3$).
- **Primary Biophysical Screen:** Surface Plasmon Resonance (SPR) utilizing biotinylated recombinant human IL-23 immobilized on a streptavidin sensor chip. Direct binding kinetics are monitored to identify fragments with $K_D \leq 500$ μ M.
- **Orthogonal Validation:** Saturation Transfer Difference NMR (STD-NMR) and 19 F-NMR using fluorinated fragment libraries.
- **Structural Elucidation:** X-ray crystallographic soaking into IL-23 crystals pre-formed in the induced-pocket conformation (leveraging crystallization conditions from PDB 8UUI). This enables structure-guided fragment growing and fragment linking across subpockets S1, S2, and S3.

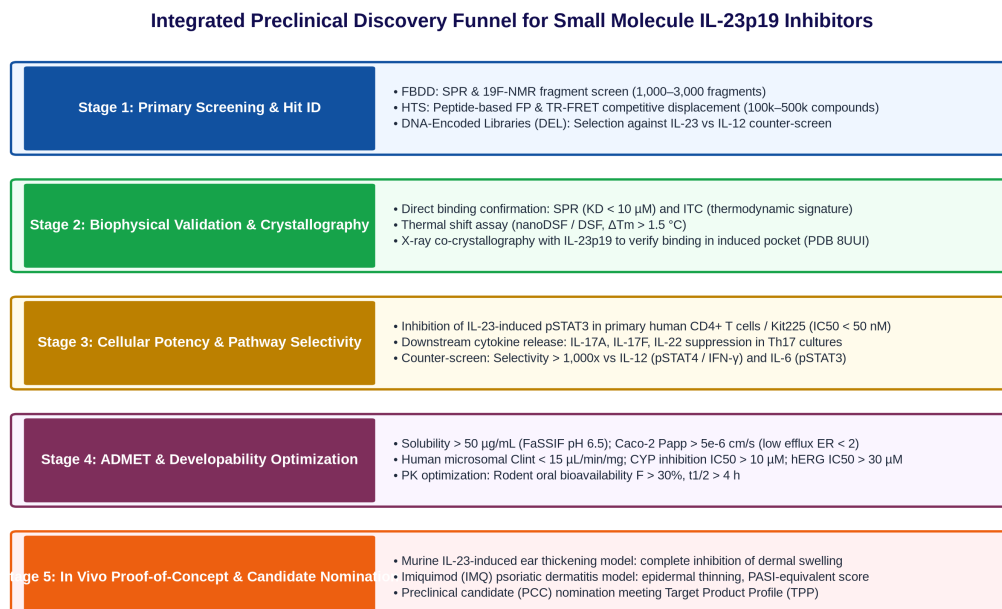


Figure 6. Integrated preclinical discovery funnel for small-molecule IL-23p19 / IL-23R orthosteric inhibitors. The cascade progresses systematically from primary hit identification (FBDD, HTS, DEL) through biophysical confirmation, in vitro cellular potency and pathway counter-screening, multi-parameter ADMET optimization, to in vivo proof-of-concept in translational psoriasis models.

4.1.2 High-Throughput Screening (HTS) via Homogeneous Probe Displacement

The discovery of optimized peptide derivatives of 23-446 provides a validated chemical probe for competitive displacement HTS [9].

- **Fluorescence Polarization (FP) Assay:** A high-affinity peptide tracer conjugated to a fluorophore (e.g., fluorescein or Cy5) exhibits high polarization when bound to IL-23p19 ($K_D \sim 1 - 10 \text{ nM}$). Small molecules displacing the tracer induce a dose-dependent decrease in fluorescence polarization.
- **TR-FRET Alternative:** Terbium-labeled anti-p40 or anti-p19 antibody donor combined with a fluorescently labeled peptide acceptor provides a robust, ratiometric readout with high signal-to-noise ($Z' > 0.75$), resistant to compound autofluorescence.
- **Library Execution:** Screening diverse chemical collections (200 000–500 000 compounds) at $10 \mu\text{M}$.

4.1.3 DNA-Encoded Chemical Libraries (DEL)

DEL technology enables the affinity selection of billions of chemical structures against immobilized target proteins.

- **Selection Architecture:** Biotinylated human IL-23 (p19/p40) is immobilized on magnetic streptavidin beads.
- **Negative Counter-Selection:** Parallel or prior selection against recombinant human IL-12 (p35/p40) is essential to subtract all chemical tags binding to the shared p40 subunit.

- **Competitive Elution:** Bound DEL members can be selectively eluted using high concentrations of untagged peptide 23-446, ensuring isolation of compounds that bind specifically to the induced orthosteric pocket.
- **Hit Confirmation:** Top-enriched sequence families are resynthesized off-DNA and confirmed by SPR and ITC.

4.1.4 Structure-Based Virtual Screening (SBVS) and AI Generative Design

High-resolution coordinates of the induced pocket (PDB 8UUI) allow virtual screening of vast make-on-demand chemical spaces (e.g., Enamine REAL, > 6 billion molecules). Computational docking, followed by molecular dynamics (MD) relaxation and MM-GBSA rescoring, enriches for scaffolds presenting optimal shape complementarity to subpockets S1 and S3.

Table 4. Comparison of Small-Molecule Discovery Strategies for the IL-23p19 Induced Pocket.

Strategy	Primary Technology	Key Advantages	Primary Challenges
FBDD	SPR, ¹⁹ F-NMR, X-ray soaking	High ligand efficiency; maps subpockets directly	Low starting affinity; requires high-throughput crystallography
HTS (FP/TR-FRET)	Peptide tracer competitive displacement	Rapid screening of > 300k compounds; high Z'	Potential false positives from aggregator or fluorescent compounds
DEL Selection	Recombinant IL-23 vs. IL-12 counter-selection	Enormous chemical space (> 10 ⁹ cpds); selection efficiency	Off-DNA translation risk; potential binding to DNA tag
SBVS & AI Design	Ensemble docking, generative pharmacophore models	Ultra-low cost; searches billions of on-demand molecules	Accuracy depends on scoring function; induced fit flexibility
Receptor peptidization	De-Peptidomimetic evolution from icotrokinra / PTG-200	Validated biological target; known clinical pharmacophore	Difficult to contract large macrocycle (> 1.5 kDa) into small molecule

5 Pharmacophore Architecture and Medicinal Chemistry Blueprint

To translate crystallographic observations into drug-like leads, the induced orthosteric pocket on IL-23p19 can be deconvoluted into four distinct functional subpockets (Fig. 7).

5.1 Subpocket Definition and Structural Requirements

1. Subpocket 1 (S1, Deep Lipophilic Anchor):

- *Structural Lining:* Pro113, Leu115, Leu116, and Leu159.
- *Pharmacophore Role:* Mimics the indole side chain of Trp18.
- *Preferred Chemotypes:* Bicyclic heteroaromatic ring systems such as indole, 5-fluoroindole, indazole, benzimidazole, quinoline, or benzoxazole. The heteroatom NH can form an ancillary hydrogen bond with the backbone carbonyl of Leu115.

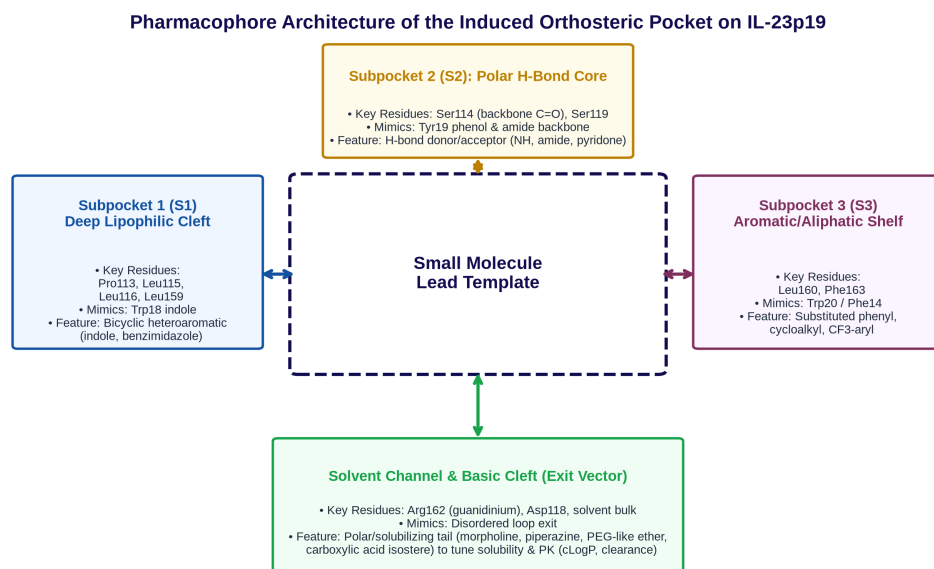


Figure 7. Four-subpocket pharmacophore architecture of the induced orthosteric pocket on IL-23p19 derived from PDB 8UUI. Subpocket 1 (S1) accommodates a bicyclic lipophilic anchor mimicking Trp18. Subpocket 2 (S2) captures essential hydrogen-bonding interactions with Ser114 and Ser119 mimicking Tyr19. Subpocket 3 (S3) provides an aromatic shelf lined by Leu160 and Phe163. The solvent exit channel (S4) allows incorporation of solubilizing and pharmacokinetic-tuning groups directed toward Arg162 and bulk solvent.

2. Subpocket 2 (S2, Polar Hydrogen-Bonding Core):

- *Structural Lining*: Backbone carbonyl oxygen of Ser114 and side-chain hydroxyl of Ser119.
- *Pharmacophore Role*: Mimics the key recognition contacts of Tyr19.
- *Preferred Chemotypes*: Hydrogen-bond donor/acceptor moieties including 2-pyridones, cyclic ureas, oxazoles, or primary carboxamides. An amide NH positioned to donate to Ser114 ($d \approx 2.7 - 2.9 \text{ \AA}$) is critical for achieving nanomolar affinity.

3. Subpocket 3 (S3, Secondary Aromatic / Aliphatic Shelf):

- *Structural Lining*: Leu160 and Phe163.
- *Pharmacophore Role*: Mimics the hydrophobic packing of Phe14 and Trp20.
- *Preferred Chemotypes*: Monocyclic substituted aryl or heteroaryl groups (e.g., 4-chlorophenyl, 3,4-difluorophenyl, 5-chloropyridin-2-yl) or rigid cycloalkyl rings (e.g., cyclopropyl, cyclohexyl, bicyclo[1.1.1]pentyl).

4. Subpocket 4 (S4, Solvent Channel and Solubilization Vector):

- *Structural Lining*: Arg162 guanidinium group, Asp118 carboxylate, and bulk solvent.
- *Pharmacophore Role*: Vector projecting out of the binding pocket into solvent, enabling fine-tuning of ADME properties without perturbing core affinity.
- *Preferred Chemotypes*: Solubilizing basic or polar appendages such as morpholine, *N*-methylpiperazine, azetidin-3-ol, piperidine, or glycol ethers. Basic nitrogen atoms can engage in a favorable long-range electrostatic interaction with Asp118 while avoiding repulsive clashes with Arg162.

5.2 Medicinal Chemistry Property Guidelines

Compounds designed against this pharmacophore should adhere to the following multi-parameter optimization (MPO) boundaries:

- **Molecular Weight (MW):** $400 \text{ Da} \leq \text{MW} \leq 550 \text{ Da}$ (or $\leq 700 \text{ Da}$ for rigid macrocycles).
- **Lipophilicity (cLogP):** $2.0 \leq \text{cLogP} \leq 3.5$, avoiding excessive lipophilicity that promotes high plasma protein binding, rapid metabolic clearance, and hERG inhibition.
- **Polar Surface Area (tPSA):** $70 \text{ \AA}^2 \leq \text{tPSA} \leq 120 \text{ \AA}^2$, balancing passive transcellular gut permeability with thermodynamic solubility.
- **Rotatable Bonds (RotB):** $\text{RotB} \leq 7$, minimizing conformational entropy penalties upon pocket binding.

6 Target Product Profile (TPP) for an Oral Small-Molecule Inhibitor

A comprehensive Target Product Profile (TPP) is detailed in [Table 5](#), outlining the minimal acceptable and optimal target criteria for an oral small-molecule development candidate for plaque psoriasis.

Table 5. Preclinical Target Product Profile (TPP) for an Oral Small-Molecule IL-23p19 Inhibitor.

Parameter	Minimal Profile	Acceptable	Optimal Target Profile	Testing Method / Standard
1. Primary Indication and Target Biology				
Primary Indication	Moderate-to-severe plaque psoriasis		Plaque psoriasis, PsA, and IBD (Crohn's, UC)	Clinical positioning
Target / MoA	Selective orthosteric inhibition of IL-23p19 / IL-23R interaction		Reversible, non-covalent, selective IL-23p19 pocket binder	Biochemical & structural confirmation (X-ray)
2. Biochemical and Cellular Potency				
Binding Affinity (K_D)	$K_D \leq 25 \text{ nM}$		$K_D \leq 1.0 \text{ nM}$	Surface Plasmon Resonance (SPR)
pSTAT3 Cellular IC_{50}	$\text{IC}_{50} \leq 50 \text{ nM}$		$\text{IC}_{50} \leq 5.0 \text{ nM}$	IL-23 stimulated human CD4+ T cells / Kit225
Cytokine Release	$\text{IC}_{50} \leq 100 \text{ nM}$ for IL-17A		$\text{IC}_{50} \leq 10 \text{ nM}$ for IL-17A/F	Primary human Th17 cell culture
Pathway Selectivity	> 500-fold vs. IL-12; > 100-fold vs. IL-6		> 2,000-fold vs. IL-12, IL-6, IL-27, IL-35	Counter-screens (pSTAT4 / IFN- γ ; pSTAT3)
3. Physicochemical Properties				
Molecular Weight	$\leq 600 \text{ Da}$		$\leq 500 \text{ Da}$	Mass spectrometry
cLogP / LogD (pH 7.4)	$\text{cLogP} \leq 4.0$; $\text{LogD} \leq 3.5$		$2.0 \leq \text{cLogP} \leq 3.0$; $\text{LogD} \leq 2.5$	Shake-flask / HPLC retention
Aqueous Solubility	$> 30 \mu\text{g mL}^{-1}$ at pH 6.5		$> 100 \mu\text{g mL}^{-1}$ in FaSSIF	Kinetic & thermodynamic solubility

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Parameter	Minimal Profile	Acceptable	Optimal Target Profile	Testing Method / Standard
Permeability	$P_{app} \geq 3 \times 10^{-6}$ cm/s		$P_{app} \geq 10 \times 10^{-6}$ cm/s	Caco-2 transwell (A-to-B)
Efflux Ratio	Efflux Ratio < 2.5		Efflux Ratio < 1.5	Caco-2 with P-gp / BCRP inhibitors
4. In Vitro ADME and Metabolism				
Microsomal Stability	$t_{1/2} > 30$ min (human)		$t_{1/2} > 60$ min (human, rat, dog)	Human/rat liver microsomes
Intrinsic Clearance	$CL_{int} < 25 \mu\text{L min}^{-1} \text{mg}^{-1}$	<	$CL_{int} < 10 \mu\text{L min}^{-1} \text{mg}^{-1}$	Human hepatocytes / microsomes
Plasma Protein Binding	Unbound fraction $f_u > 0.01$		Unbound fraction $f_u > 0.05$	Rapid equilibrium dialysis (RED)
CYP Inhibition	$IC_{50} > 10 \mu\text{M}$ across major CYPs		$IC_{50} > 30 \mu\text{M}$ across panel; no TDI	Recombinant CYP luminescence
CYP Induction	< 2-fold induction at $10 \mu\text{M}$		No significant induction (mRNA)	Cryopreserved human hepatocytes
5. Pharmacokinetics and In Vivo Exposure				
Oral Bioavailability (F)	$F > 25\%$ (rodent, dog)		$F > 50\%$ (rodent, cynomolgus)	Pharmacokinetic studies in vivo
Elimination Half-Life ($t_{1/2}$)	$t_{1/2} \geq 4$ h in non-rodent		$t_{1/2} \geq 8$ h (supports QD dosing)	In vivo IV / PO PK profiling
Volume of Distribution	$V_{ss} > 0.7 \text{ L kg}^{-1}$		$V_{ss} \approx 1.0 - 2.5 \text{ L kg}^{-1}$	Tissue distribution into skin
6. Safety, Toxicology, and Developability				
hERG Cardiac Liability	$IC_{50} > 10 \mu\text{M}$; margin > 30×		$IC_{50} > 30 \mu\text{M}$; margin > 100×	Automated patch clamp
Genotoxicity	Negative in Ames test		Negative in Ames and in vitro MNT	OECD 471 and 487 assays
Cytotoxicity	$CC_{50} > 50 \mu\text{M}$		$CC_{50} > 100 \mu\text{M}$ in HepG2	Neutral red / CellTiter-Glo
14-Day In Vivo Tolerability	NOAEL > 10× therapeutic exposure		NOAEL > 30× therapeutic exposure	Rodent dose-range finding (DRF)
7. In Vivo Efficacy in Preclinical Models				
IL-23 Ear Swelling Model	$\geq 70\%$ inhibition of swelling		$\geq 90\%$ inhibition (comparable to mAb)	Intradermal IL-23 injection in mice
Imiquimod Psoriasis Model	$\geq 60\%$ reduction in epidermal thickness		$\geq 80\%$ reduction; normalization of histology	7-day IMQ topical application
Translational Biomarkers	$\geq 70\%$ reduction in skin pSTAT3		Complete suppression of cutaneous IL-17A/F	qPCR, Western blot, IHC

7 Preclinical Testing Cascade and Translational Biomarkers

To ensure seamless triage of screening hits toward candidate nomination, a tiered testing cascade must be established:

7.1 Tiered Testing Workflow

1. Tier 1: Primary Hit Triaging (Biochemical / Biophysical)

- FP or TR-FRET probe displacement assay ($n = 2$, 10-point IC_{50} determination).
- SPR direct binding (K_D , k_{on} , k_{off} on immobilized IL-23p19/p40).
- Counter-screen on immobilized IL-12p35/p40 to eliminate p40-directed hits.
- Deliverable: Confirmed p19-selective binders with $K_D < 1 \mu M$.

2. Tier 2: Orthogonal Biophysical and Structural Characterization

- Isothermal Titration Calorimetry (ITC) to confirm 1:1 binding stoichiometry and thermodynamic profile (ΔH , $-T\Delta S$).
- Differential Scanning Fluorimetry (nanoDSF) measuring target thermal stabilization ($\Delta T_m > 1.5^\circ C$).
- X-ray co-crystallography of IL-23 in complex with lead series to confirm pose in the induced orthosteric pocket.

3. Tier 3: In Vitro Functional and Pathway Selectivity Profiling

- Primary human CD4+ T cells stimulated with recombinant IL-23 (10 ng mL^{-1}); quantification of pSTAT3 by homogeneous HTRF or flow cytometry ($IC_{50} < 50 \text{ nM}$).
- Primary human Th17 differentiation cultures: dose-dependent inhibition of IL-17A, IL-17F, and IL-22 secretion by ELISA / multiplex MSD.
- Selectivity counter-screens:
 - (a) IL-12 stimulated human PBMCs (pSTAT4 and IFN- γ release; required selectivity > 1000 -fold).
 - (b) IL-6 stimulated human PBMCs (pSTAT3 release; required selectivity > 1000 -fold).

4. Tier 4: In Vitro ADMET and Physicochemical Profiling

- Thermodynamic solubility in FaSSIF/FeSSIF (pH 6.5).
- Caco-2 bidirectional permeability and efflux ratio (P-gp / BCRP substrate evaluation).
- Phase I metabolic stability (human, rat, mouse, dog liver microsomes and hepatocytes).
- CYP inhibition (1A2, 2C9, 2C19, 2D6, 3A4) and time-dependent inhibition (TDI).
- In vitro cardiac liability: automated hERG patch-clamp ($IC_{50} > 30 \mu M$).
- In vitro genotoxicity: Ames bacterial reverse mutation test and micronucleus test (MNT).

5. Tier 5: In Vivo Pharmacokinetics, Efficacy, and Biomarkers

- In vivo PK in rodents (IV and oral dosing at 2 mg kg^{-1} and 10 mg kg^{-1}); determination of clearance, V_{ss} , oral bioavailability (F), and skin tissue penetration ($K_{p,skin} \geq 1.0$).
- Acute Pharmacodynamic Model: Recombinant mouse/human IL-23 intradermal injection into mouse ears for 5 consecutive days. Readout: ear thickness measurement and cutaneous IL-17A/F and pSTAT3 levels.
- Chronic Translational Model: Imiquimod (IMQ)-induced psoriatic skin inflammation in C57BL/6 mice (daily topical 62.5 mg IMQ cream for 7 days). Oral administration of candidate compound QD/BID. Readout: clinical erythema, scaling, and thickness scoring, histological epidermal thickness, and cutaneous cytokine/chemokine profiling.

7.2 Translational Biomarkers for Human Clinical Transition

A critical advantage of targeting the IL-23p19 / IL-23R axis is the rich repertoire of clinically validated translational biomarkers:

- **Circulating Biomarkers:** Serum levels of IL-17A, IL-17F, and IL-22 provide direct systemic readouts of pathway inhibition.
- **Cutaneous Tissue Biomarkers:** Quantitative RT-PCR and immunohistochemistry (IHC) on punch biopsies from psoriatic lesional skin evaluate downregulation of downstream keratinocyte activation markers: psoriasin (S100A7), calgranulin A/B (S100A8/S100A9), human β -defensin 2 (DEFB4), and lipocalin-2 (LCN2).
- **Intracellular Pathway Readout:** Phosphorylated STAT3 (pSTAT3 Tyr705) in circulating memory T cells and skin biopsy sections directly monitors target engagement.

8 Conclusions and Strategic Preclinical Roadmap

The IL-23p19 / IL-23R protein-protein interface represents one of the most therapeutically validated targets in contemporary immunology. While monoclonal antibodies have set an exceptionally high bar for clinical clearance in plaque psoriasis, their injectable route of administration, immunogenicity, and high cost leave substantial room for an oral small-molecule alternative.

The recent identification and crystallographic validation of an induced-fit orthosteric pocket on IL-23p19 (PDB [8UUI](#)) fundamentally transforms the druggability assessment of this target:

1. **Druggable Cavity Formation:** The dynamic displacement of loops 108–122 and 153–165 (with Ca shifts up to 3.93 Å for Pro155) creates a well-defined lipophilic cleft ($\sim 450 \text{ \AA}^3$) suitable for rule-of-5-compliant small-molecule ligands.
2. **Clear Pharmacophore Blueprint:** The four-subpocket model (S1 lipophilic core, S2 hydrogen-bonding recognition, S3 aromatic shelf, and S4 solvent exit vector) provides explicit medicinal chemistry guidance for hit-to-lead and lead-optimization campaigns.
3. **Actionable Discovery Funnel:** Synergy between fragment-based screening (FBDD), peptide-displacement HTS assays, and DNA-encoded library (DEL) counter-selection on IL-23 versus IL-12 offers multiple parallel avenues to generate high-affinity chemical starting points.
4. **Stringent TPP:** Adherence to the established Target Product Profile will ensure that preclinical development candidates deliver biologic-like efficacy ($\text{PASI-75} \geq 75\%$, $\text{PASI-90} \geq 55\%$) via convenient once-daily oral administration, opening a transformative new chapter in psoriasis therapeutics.

Recommended 18-Month Preclinical Roadmap:

- **Months 1–4 (Hit Identification):** Execute dual HTS (peptide FP probe displacement, 300k compounds) and DEL selection (biotinylated IL-23 vs. IL-12 counter-selection); initiate FBDD SPR screen (2,000 fragments).
- **Months 5–8 (Hit-to-Lead):** Biophysical confirmation by ITC and nanoDSF; establish X-ray co-crystallography in the induced-pocket form; demonstrate primary cellular pSTAT3 inhibition ($\text{IC}_{50} < 200 \text{ nM}$).

- **Months 9–14 (Lead Optimization):** Structure-based elaboration of S1, S2, and S3 subpockets; incorporate S4 solubilizing groups to optimize Caco-2 permeability ($P_{app} > 5 \times 10^{-6}$ cm/s) and microsomal stability ($CL_{int} < 15 \mu\text{L min}^{-1} \text{mg}^{-1}$); advance oral bioavailability in rodents ($F > 30\%$).
- **Months 15–18 (Candidate Nomination):** Demonstrate in vivo efficacy in murine IL-23-induced ear swelling and IMQ psoriatic dermatitis models; complete 14-day rodent DRF tolerability and in vitro safety panels; nominate Preclinical Development Candidate (PCC) satisfying all TPP criteria.

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